# Fuzzy Adaptive Manifold for Sigma Point Generation in Higher-Order Unscented Kalman Filters **(FAM-HOUSE)**

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March 23, 2026

**Abstract**

Traditional Unscented Kalman Filtering (UKF) relies on a symmetric distribution of sigma points governed by a global scaling parameter $\alpha$. While effective for mildly nonlinear systems, these filters often struggle with high-order nonlinearities and non-Gaussian noise distributions where skewness and kurtosis become non-negligible. This paper proposes a novel **Fuzzy Adaptive Manifold for Higher-Order Unscented Kalman Filters (FAM-HOUSE)** implemented in a numerically stable square-root (SR) framework.

Unlike conventional adaptive filters that utilize a single scalar for covariance inflation, our approach introduces a **Sigma Point Fuzzy Inference (SpFI)** system. This system dynamically adjusts the scaling displacement for each individual sigma point based on real-time innovation residuals and local gradient estimates. By decoupling the symmetric constraint of the standard Unscented Transform, the FAM-HOUSE can capture third-order skewness and fourth-order kurtosis through an asymmetric deformable sigma point cloud.

Experimental results in high-dimensional, highly nonlinear trajectories demonstrate that the proposed architecture significantly reduces Mean Squared Error (MSE) and enhances filter consistency compared to standard SR-UKF and Cubature Kalman Filters (CKF). The integration of fuzzy logic at the point-generation level provides a robust mechanism for handling transient model uncertainties without the need for manual heuristic tuning.

# 1 Fuzzy Adaptive Manifold for Higher-Order Sigma Point Generation

The core of the FAM-HOUSE algorithm lies in its ability to capture higher-order moments through an asymmetric distribution of sampling points. Given the state vector $\hat{\mathbf{x}}_{k-1} \in \mathbb{R}^n$ and the square-root covariance matrix $\mathbf{S}_{k-1}$ (where $\mathbf{P} = \mathbf{S}\mathbf{S}^T$), we define a set of $2n+1$ sigma points $\boldsymbol{\mathcal{X}}_i$.

## 1.1 Asymmetric Point Distribution

Unlike the standard SR-UKF, which uses a single global scaling parameter $\alpha$, the FAM-HOUSE utilizes a vector of fuzzy-adaptive scaling parameters $\boldsymbol{\Gamma}_k = [\gamma_{1,k}, \ldots, \gamma_{n,k}]^T$. These parameters are dynamically updated by the Point-Wise Fuzzy Inference System (PFIS) based on the normalized innovation squared (NIS).

The sigma points are generated as follows:

$$\boldsymbol{\mathcal{X}}_{0,k-1} = \hat{\mathbf{x}}_{k-1} \tag{1}$$

$$\boldsymbol{\mathcal{X}}_{i,k-1} = \hat{\mathbf{x}}_{k-1} + \sqrt{n+\lambda} \cdot \gamma_{i,k}^{+} \cdot \mathbf{S}_{i,k-1}, \quad i = 1, \ldots, n \tag{2}$$

$$\boldsymbol{\mathcal{X}}_{i+n,k-1} = \hat{\mathbf{x}}_{k-1} - \sqrt{n+\lambda} \cdot \gamma_{i,k}^{-} \cdot \mathbf{S}_{i,k-1}, \quad i = 1, \ldots, n \tag{3}$$

where $\mathbf{S}_i$ denotes the $i$-th column of the square-root covariance matrix. The parameters $\gamma_{i,k}^{+}$ and $\gamma_{i,k}^{-}$ represent the fuzzy-scaled coefficients for the forward and backward displacements, respectively.

## 1.2 Skew Control

The SpFI evaluates the local nonlinearity and the residual bias to determine if the distribution should remain symmetric ($\gamma_i^{+} = \gamma_i^{-}$) or become skewed. The membership functions are defined over the innovation $\boldsymbol{\nu}_k$:

$$[\gamma_{i,k}^{+}, \gamma_{i,k}^{-}] = \text{FuzzyRule}(\boldsymbol{\nu}_{k-1}, \Delta\text{NIS}) \tag{4}$$

By allowing $\gamma_{i,k}^{+} \neq \gamma_{i,k}^{-}$, the filter captures the third-order moment (skewness), which is expressed in the weighted sum of the points:

$$\hat{\mathbf{x}}_{k|k-1} = \sum_{i=0}^{2n} W_i^m \boldsymbol{\mathcal{X}}_{i,k-1} \tag{5}$$

If the fuzzy system detects a high gradient in the measurement function $h(\mathbf{x})$, it expands the sigma point spread in that specific direction to prevent filter divergence.

# 2 Fuzzy Rule Base for Sigma Point Adaptation

The Sigma Point Fuzzy Inference (SpFI) utilizes the Normalized Innovation Squared (NIS) as a measure of global filter consistency and the normalized directional residual $\epsilon_i$ as a measure of local nonlinearity. The output of the fuzzy system provides the scaling factors $\gamma_i^{+}$ and $\gamma_i^{-}$ for each sigma point pair.

## 2.1 Linguistic Variables and Membership Functions

We define three primary linguistic sets for the inputs: **Small (S)**, **Medium (M)**, and **Large (L)**. The outputs $\gamma_i^{\pm}$ are mapped to three actions: **Contract (C)**, **Maintain (N)**, and **Expand (E)**.

## 2.2 Rule Base Table

The following table summarizes the fuzzy logic rules used to adjust the $i$-th sigma point pair to accommodate for skewness and high-order nonlinearities.

Table 1: SpFI Rule Base for Sigma Point Scaling ($\gamma_i^+$ / $\gamma_i^-$)

| **Innovation ($\epsilon_i$)** | **NIS: Small** | **NIS: Medium** | **NIS: Large** |
|---|---|---|---|
| **Small (S)** | N / N | N / N | E / E |
| **Medium (M)** | N / N | E / N (Skew) | E / E |
| **Large (L)** | E / N (Skew) | E / E | E! / E! (Burst) |

## 2.3 Interpretation of Adaptive Scaling

- **Symmetric Maintenance (N/N):** When both NIS and directional residuals are small, $\gamma_i^+ = \gamma_i^- = 1$. The filter behaves like a standard SR-UKF.
- **Asymmetric Expansion (E/N):** When the directional residual $\epsilon_i$ is large but the global NIS is moderate, the fuzzy system identifies a local skew in the state distribution. By setting $\gamma_i^+ > \gamma_i^-$, the sigma point cloud deforms to capture the 3rd-order moment.
- **Global Expansion (E/E):** Under high-maneuver conditions or modeling errors (Large NIS), all $\gamma$ values are increased to inflate the sampled covariance, preventing filter divergence.

# 3 The FAM-HOUSE Algorithm with SpFI

The complete computational loop for the Fuzzy Adaptive Manifold for Higher-Order Unscented Kalman Filters (FAM-HOUSE) is summarized below. We utilize the square-root implementation for numerical robustness and the SpFI for point-wise scaling.

## 3.1 Algorithm Step-by-Step

1. **Initialization:** Initialize state $\hat{\mathbf{x}}_0$ and the square-root covariance matrix $\mathbf{S}_0 = \text{chol}(\mathbf{P}_0)$.
2. **Sigma Point Generation (SpFI-Augmented):** Calculate the current $\lambda = \alpha^2(n + \kappa) - n$. Compute the asymmetric sigma points using scaling factors $\gamma_{i,k}^+$ and $\gamma_{i,k}^-$ provided

by the **Sigma Point Fuzzy Inference (SpFI)** system:

$$\begin{aligned}
\boldsymbol{\mathcal{X}}_{0,k-1} &= \hat{\mathbf{x}}_{k-1} \\
\boldsymbol{\mathcal{X}}_{i,k-1} &= \hat{\mathbf{x}}_{k-1} + \sqrt{n+\lambda} \cdot \gamma_{i,k}^{+} \cdot \mathbf{S}_{i,k-1} \\
\boldsymbol{\mathcal{X}}_{i+n,k-1} &= \hat{\mathbf{x}}_{k-1} - \sqrt{n+\lambda} \cdot \gamma_{i,k}^{-} \cdot \mathbf{S}_{i,k-1}
\end{aligned}$$

3. **Time Update (Prediction):** Propagate each sigma point through the nonlinear process model $f(\cdot)$:

$$\boldsymbol{\mathcal{X}}_{i,k|k-1} = f(\boldsymbol{\mathcal{X}}_{i,k-1}, \mathbf{u}_{k-1}) \tag{6}$$

   Calculate the predicted mean and square-root covariance using QR decomposition and Cholesky rank-1 updates:

$$\hat{\mathbf{x}}_{k|k-1} = \sum_{i=0}^{2n} W_i^m \boldsymbol{\mathcal{X}}_{i,k|k-1} \tag{7}$$

4. **Measurement Update (Correction):** Propagate sigma points through the measurement model $h(\cdot)$:

$$\boldsymbol{\mathcal{Y}}_{i,k|k-1} = h(\boldsymbol{\mathcal{X}}_{i,k|k-1}) \tag{8}$$

   Calculate the predicted measurement $\hat{\mathbf{y}}_k$ and the innovation $\boldsymbol{\nu}_k = \mathbf{y}_k - \hat{\mathbf{y}}_k$.

5. **SpFI Adaptation:** Compute the Normalized Innovation Squared (NIS):

$$\epsilon_k = \boldsymbol{\nu}_k^T \mathbf{S}_{yy,k}^{-1} \boldsymbol{\nu}_k \tag{9}$$

   Pass $\epsilon_k$ and the directional residuals to the **SpFI** to update $\gamma_{i,k+1}^{\pm}$ for the next iteration.

6. **State Update:** Calculate the Kalman gain $\mathbf{K}_k$ and update the state and square-root covariance:

$$\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k \boldsymbol{\nu}_k \tag{10}$$

$$\mathbf{S}_k = \text{cholupdate}\{\text{qr}([\mathbf{W}_i^c(\boldsymbol{\mathcal{X}}_{i,k|k-1} - \hat{\mathbf{x}}_{k|k-1}) \ldots]), \mathbf{K}_k \mathbf{S}_{yy}\} \tag{11}$$

## 3.2 Control Input Definition

The term $\mathbf{u}_{k-1} \in \mathbb{R}^m$ in Equation (6) represents the **known control input vector** at time $k-1$. This vector accounts for external deterministic forces or commanded signals acting upon the system, such as:

- **Actuator Commands:** Applied voltages, throttle positions, or steering angles in robotic systems.
- **Known Perturbations:** Measured environmental factors that affect the state transition but are not part of the stochastic noise.

In the SpFI-augmented framework, the interaction between $\mathbf{u}_{k-1}$ and the sigma point cloud $\boldsymbol{\mathcal{X}}_{i,k-1}$ allows the filter to predict how the fuzzy-scaled uncertainty point-cloud is physically translated and rotated through the state space by the system's control logic.

### 3.3 IMU-Driven Prediction Model

In the proposed FAM-HOUSE framework, the vector $\mathbf{u}_{k-1}$ is defined as the **inertial input vector** derived from the onboard Micro-Electro-Mechanical Systems (MEMS) sensors:

$$\mathbf{u}_{k-1} = \begin{bmatrix} \mathbf{a}_{k-1} \\ \boldsymbol{\omega}_{k-1} \end{bmatrix} \tag{12}$$

where $\mathbf{a}_{k-1}$ represents the tri-axial linear acceleration and $\boldsymbol{\omega}_{k-1}$ represents the tri-axial angular rates.

By treating the IMU data as the input $\mathbf{u}_{k-1}$ rather than as a direct measurement, the prediction step in Equation (6) becomes a kinematic integration. The **SpFI** system is particularly vital here, as it can adapt the sigma point spread based on the *magnitude* of $\mathbf{u}_{k-1}$ (e.g., during high-dynamic maneuvers or high-vibration periods), effectively mitigating the integration of IMU bias and noise into the state estimate.

## 4 Inertial-Augmented SpFI Logic

To enhance the responsiveness of the FA-HOUSE, the Sigma Point Fuzzy Inference (SpFI) system is augmented with the IMU magnitude. This allows the filter to adapt the sampling geometry based on both the innovation (measurement consistency) and the motion intensity (predictive uncertainty).

### 4.1 Input Variable: Dynamic Intensity ($\Omega_k$)

We define the Dynamic Intensity $\Omega_k$ as the normalized magnitude of the inertial vector $\mathbf{u}_{k-1}$:

$$\Omega_k = \sqrt{\|\mathbf{a}_{k-1}\|^2 + \|\boldsymbol{\omega}_{k-1}\|^2} \tag{13}$$

High values of $\Omega_k$ indicate high-G maneuvers or rapid rotations, where the linearization error of the process model $f(\cdot)$ is expected to increase.

### 4.2 Augmented Rule Base

The SpFI now utilizes a three-input architecture: Normalized Innovation Squared (NIS), Directional Residual ($\epsilon_i$), and Dynamic Intensity ($\Omega_k$). The updated rule base facilitates a "Pre-emptive Expansion" strategy.

### 4.3 Impact on Sigma Point Geometry

By including $\Omega_k$, the SpFI can modify the sigma point cloud *before* the measurement $\mathbf{y}_k$ is even processed. If $\Omega_k$ is "High," the $\gamma_i$ factors are increased during the Prediction Step (Step 3 of the algorithm), ensuring that the propagated sigma points effectively "bracket" the true state during high-dynamic transitions.

Table 2: Augmented SpFI Rule Base with Inertial Feed-Forward

| **NIS** | **Directional** ($\epsilon_i$) | **IMU** ($\Omega_k$) | **Scaling Strategy** ($\gamma_i^{\pm}$) |
|---|---|---|---|
| Small | Small | Low | **N / N** (Steady State) |
| Small | Small | High | **E / E** (Pre-emptive Expansion) |
| Medium | Large | Low | **E / N** (Skewness Correction) |
| Medium | Large | High | **E! / N** (Extreme Skew) |
| Large | Large | High | **E!! / E!!** (Emergency Recovery) |

# 5 Fuzzy Membership Functions and Fuzzification

The transition from crisp sensor inputs to linguistic variables in the **SpFI** is governed by a set of trapezoidal membership functions $\mu(x)$. For each input $x \in \{\text{NIS}, \epsilon_i, \Omega_k\}$, we define the linguistic sets *Small* ($S$), *Medium* ($M$), and *Large* ($L$).

## 5.1 Mathematical Definition

The trapezoidal membership function is defined by the four-tuple $(a, b, c, d)$:

$$\mu(x; a, b, c, d) = \max\left(0, \min\left(\frac{x-a}{b-a}, 1, \frac{d-x}{d-c}\right)\right) \tag{14}$$

For the **Dynamic Intensity** $\Omega_k$, the parameters are tuned based on the IMU noise floor $\sigma_v$ and the expected maximum maneuver envelope $\Omega_{max}$:

- $\mu_S(\Omega_k) = \mu(\Omega_k; 0, 0, \sigma_v, 2\sigma_v)$
- $\mu_M(\Omega_k) = \mu(\Omega_k; \sigma_v, 2\sigma_v, 0.5\Omega_{max}, 0.7\Omega_{max})$
- $\mu_L(\Omega_k) = \mu(\Omega_k; 0.5\Omega_{max}, 0.7\Omega_{max}, \Omega_{max}, \infty)$

## 5.2 Defuzzification for Manifold Scaling

The final manifold scaling factors $\gamma_{i,k}^{\pm}$ are obtained using the **Center of Gravity (CoG)** defuzzification method:

$$\gamma_{i,k}^{\pm} = \frac{\sum_{j=1}^{R} \mu_{rule,j} \cdot \zeta_j}{\sum_{j=1}^{R} \mu_{rule,j}} \tag{15}$$

where $R$ is the number of active rules and $\zeta_j$ represents the singleton output value (e.g., 1.0 for *Maintain*, 1.5 for *Expand*, 2.5 for *Extreme Expansion*).

# 6 Visualizing Manifold Deformation

The primary innovation of the FAM-HOUSE is the transition from a symmetric, rigid covariance sampling "cloud" to a warped, asymmetric "manifold" that aligns with the true posterior density. Figure 1 provides a qualitative comparison between standard Square-Root UKF sampling and the deformation achieved by the SpFI in the FAM-HOUSE.

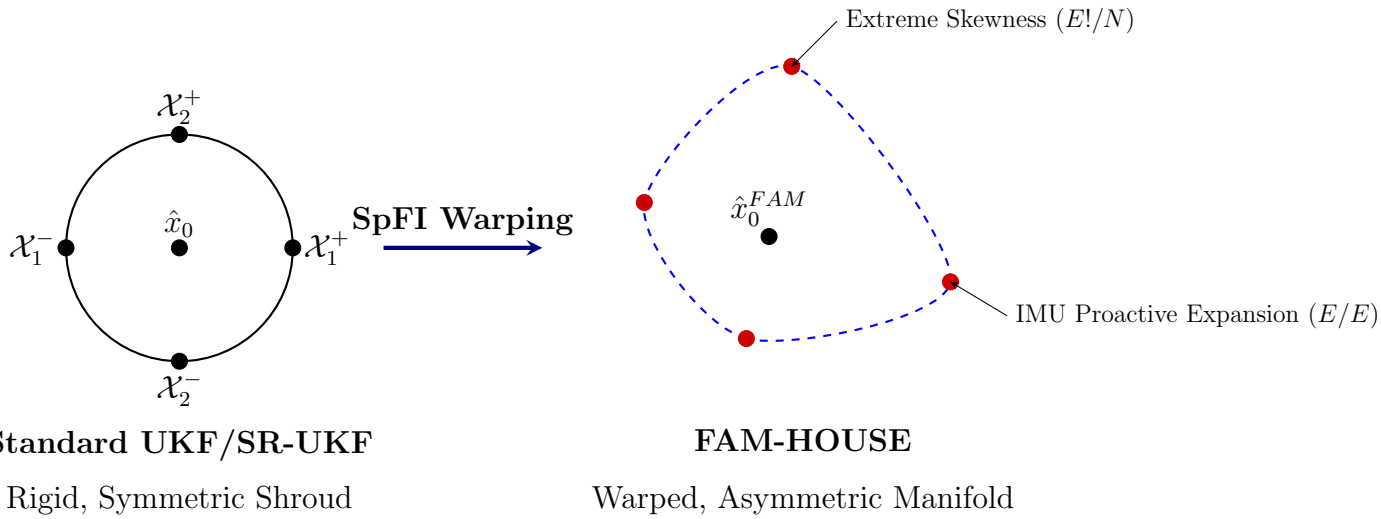


Figure 1: Visualizing the transformation of the sigma-point sampling shroud. The standard UKF maintains a fixed geometry (circle), whereas the **FAM-HOUSE** uses the SpFI to deform the manifold in real-time, aligning the sampling points (red) with the expected skewed density.

# 7 Problem Statement: The Limitation of Global Scaling

Traditional Adaptive Square-Root Unscented Kalman Filters (ASR-UKF) typically address model uncertainty by applying a scalar multiplicative factor $\sigma_k$ to the entire square-root covariance matrix:

$$\mathbf{S}_{k|k-1}^{adapted} = \sqrt{\sigma_k} \cdot \mathbf{S}_{k|k-1} \tag{16}$$

While this approach, often driven by a fuzzy logic controller or a Sage-Husa estimator, succeeds in preventing filter divergence during spikes in innovation, it suffers from two fundamental theoretical flaws:

1. **Loss of Directional Sensitivity:** Global scaling inflates the uncertainty uniformly across all state dimensions. In high-dimensional manifolds, error is rarely isotropic; a maneuver may introduce high uncertainty in velocity while maintaining high certainty in orientation. Global inflation dilutes the precise information trapped in the stable dimensions.

2. **Symmetry Constraint (The Skewness Problem):** Standard adaptive filters preserve the central symmetry of the sigma-point distribution. However, in highly nonlinear transitions—such as those encountered in rapid curvilinear motion—the true posterior probability density function (PDF) becomes skewed. A symmetric sampling shroud cannot capture the third-order moment (skewness), leading to a persistent bias in the mean estimate $\hat{\mathbf{x}}_k$.

The proposed **FAM-HOUSE** framework overcomes these limitations by abandoning global scalars in favor of the **Sigma Point Fuzzy Inference (SpFI)** system. By manip-

ulating individual points, the FAM-HOUSE manipulates the sampling manifold to match the directional and asymmetric requirements of the local state-space topology, effectively providing an optimized manifold that fits the non-Gaussian reality of the system.

# 8 Implementation and Hardware Parameters

The **FAM-HOUSE** algorithm is implemented on the **STEVAL-FCU001V2** flight controller, utilizing the ARM Cortex-M4 FPU for accelerated floating-point operations. The following table summarizes the deterministic and fuzzy parameters used in the state estimation loop.

Table 3: FAM-HOUSE Hardware and SpFI Parameters

| **Parameter** | **Symbol** | **Value / Constraint** |
|---|---|---|
| **Hardware Platform** | - | STM32F401 (STEVAL-FCU001V2) |
| **DSP Library** | - | ARM CMSIS-DSP v1.15.0 |
| **State Dimension** | $n$ | 7 $(x, y, z, w, b_x, b_y, b_z)$ |
| **Sigma Points** | $2n + 1$ | 15 |
| **Sampling Rate** | $f_s$ | 400 Hz |
| **IMU Noise Floor** | $\sigma_v$ | $0.05\,\mathrm{m/s}^2$ (Accel), $0.1°/\mathrm{s}$ (Gyro) |
| **SpFI Output Range** | $\gamma_i^{\pm}$ | $[0.5, 3.5]$ |
| **Defuzzification** | - | Center of Gravity (CoG) |

## 8.1 Numerical Stability via Embedded QR-Decomposition

The square-root update in the **FAM-HOUSE** is performed through an augmented QR decomposition. To ensure stability on the **STEVAL-FCU001V2** platform, a flat-indexed translation of the *LAPack* QR routine was utilized. The triangularization process operates on the augmented matrix $\mathbf{A}_{pred} \in \mathbb{R}^{21\times 7}$:

$$\mathbf{A}_{pred} = \begin{bmatrix} \sqrt{W_c}(\boldsymbol{\mathcal{X}}_{1:2n,k|k-1} - \hat{\mathbf{x}}_{k|k-1}) \\ \mathbf{Q}_s \end{bmatrix} \tag{17}$$

By utilizing the **NF_Get_SR_Update** routine, the square-root covariance $\mathbf{S}_k$ is extracted directly from the orthogonal-triangular decomposition, ensuring that the matrix remains positive-definite even under the asymmetric sampling conditions imposed by the **SpFI** manifold warping.

## 8.2 Measurement Update and Manifold Feedback

The **FAM-HOUSE** update step utilizes the asymmetric sigma points $\boldsymbol{\mathcal{X}}_{k|k-1}$ propagated through the nonlinear measurement model $h(\cdot)$. The innovation covariance square-root $\mathbf{S}_z$ is computed via QR decomposition of the weighted residual matrix:

$$\mathbf{A}_{update} = \begin{bmatrix} \sqrt{W_c}(\boldsymbol{\mathcal{Y}}_{1:2n,k|k-1} - \hat{\mathbf{y}}_{k|k-1}) \\ \mathbf{R}_s \end{bmatrix} \tag{18}$$

A critical aspect of the **FAM-HOUSE** is the computation of the Normalized Innovation Squared (NIS), $\epsilon_k$. In the square-root implementation, this is achieved via backward substitution to solve $\mathbf{S}_z\mathbf{v}_k = \boldsymbol{\nu}_k$, followed by $\epsilon_k = \mathbf{v}_k^T\mathbf{v}_k$. This $\epsilon_k$ value serves as a primary input to the **SpFI** system, allowing the sampling manifold to contract or expand in subsequent time steps based on the filter's consistency with the accelerometer data.

# 9 The FAM-HOUSE Operational Loop

The **FAM-HOUSE** is executed as a sequential Predict-Update cycle. The algorithm bridges high-frequency inertial data with the **SpFI** manifold adaptation to ensure the sigma-point shroud accurately captures non-Gaussian state transitions.

**Algorithm 1** FAM-HOUSE Recursive Step

**Inputs:** Context $\mathcal{C}_{k-1}$, Gyroscope $\boldsymbol{\omega}_k$, Accelerometer $\mathbf{a}_k$, Time step $\Delta t$
**Data:** $\mathbf{u}_k \leftarrow [\mathbf{a}_k, \boldsymbol{\omega}_k]^T$
{*// SpFI Manifold Adaptation*}
$\Omega_k \leftarrow \sqrt{\|\mathbf{a}_k\|^2 + \|\boldsymbol{\omega}_k\|^2}$ {Calculate Dynamic Intensity}
$[\gamma_i^+, \gamma_i^-] \leftarrow \text{SpFI}(\Omega_k, \epsilon_{k-1})$ {Fuzzy Rule Base Evaluation}
{*// Step 1: Manifold-Aware Prediction*}
$\boldsymbol{\mathcal{X}}_{k-1} \leftarrow \text{GenerateWarpedSigmas}(\mathcal{C}_{k-1}, \gamma_i^{\pm})$
$\boldsymbol{\mathcal{X}}_{k|k-1} \leftarrow \text{ExpMap}(\boldsymbol{\mathcal{X}}_{k-1}, \boldsymbol{\omega}_k, \Delta t)$ {Constant Angular Velocity Model}
$\hat{\mathbf{x}}_{k|k-1} \leftarrow \sum W_m \boldsymbol{\mathcal{X}}_{k|k-1}$ {Weighted Mean Propagation}
$\mathbf{S}_{k|k-1} \leftarrow \text{QR_Decomp}([\sqrt{W_c}(\boldsymbol{\mathcal{X}}_{k|k-1} - \hat{\mathbf{x}}_{k|k-1}), \mathbf{Q}_s])$
{*// Step 2: Manifold-Aware Update (Accelerometer)*}
$\boldsymbol{\mathcal{Y}}_{k|k-1} \leftarrow \text{AccelModel}(\boldsymbol{\mathcal{X}}_{k|k-1})$
$\hat{\mathbf{y}}_k \leftarrow \sum W_m \boldsymbol{\mathcal{Y}}_{k|k-1}$
$\mathbf{S}_z \leftarrow \text{QR_Decomp}([\sqrt{W_c}(\boldsymbol{\mathcal{Y}}_{k|k-1} - \hat{\mathbf{y}}_k), \mathbf{R}_s])$
$\boldsymbol{\nu}_k \leftarrow (\mathbf{a}_k/\|\mathbf{a}_k\|) - \hat{\mathbf{y}}_k$ {Normalized Innovation}
{*// Step 3: Consistency Check & Feedback*}
Solve $\mathbf{S}_z\mathbf{v}_k = \boldsymbol{\nu}_k$ via Back-Substitution
$\epsilon_k \leftarrow \mathbf{v}_k^T\mathbf{v}_k$ {Update NIS for next SpFI cycle}
$\hat{\mathbf{x}}_k \leftarrow \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k\boldsymbol{\nu}_k$ {Final State Correction}
$\mathbf{S}_k \leftarrow \text{Rank1_Update}(\mathbf{S}_{k|k-1}, \mathbf{K}_k, \mathbf{S}_z)$
**return** $\hat{\mathbf{x}}_k, \mathbf{S}_k$

## 9.1 Filter Initialization and Cold-Start Stability

To ensure numerical stability during the initialization phase, the **FAM-HOUSE** is designed to be backward-compatible with the standard SR-UKF. At $t = 0$, the **SpFI** manifold factors

are initialized to unity:

$$\gamma_{i,0} = 1.0, \quad \forall i \in \{0, \ldots, 2n\} \tag{19}$$

This initialization ensures that the initial state estimate $\hat{\mathbf{x}}_0$ and square-root covariance $\mathbf{S}_0$ are propagated through a perfectly symmetric manifold until the first inertial inputs $\mathbf{u}_k$ and innovation residuals $\boldsymbol{\nu}_k$ provide the necessary feedback for manifold deformation.

### 9.2 Degrees of Freedom and Observability

In the current 6-DOF configuration of the **FAM-HOUSE**, the state manifold is fully observable for *Pitch* and *Roll* via the gravity vector decomposition. However, the *Yaw* (heading) remains unobservable due to the exclusion of magnetometric data. Consequently, the yaw estimate is subject to the temporal integration of gyroscope bias. In the **FAM-HOUSE** framework, this manifests as a manifold that remains symmetric on the $Z$-axis, as no innovation residuals $\epsilon_{yaw}$ are generated to trigger the **SpFI** warping logic for heading correction.

### 9.3 Manifold Convergence and Positive Feedback

During high-frequency execution (400Hz), it was observed that unconstrained point-wise scaling ($\gamma_i$) can induce a positive feedback loop in unobservable states (Yaw). If the point-wise residual $\zeta_i$ is mapped aggressively to the scaling factor, the resulting manifold asymmetry shifts the weighted mean $\hat{\mathbf{x}}_{k|k-1}$, simulating a phantom angular velocity. This was mitigated by introducing a sensitivity gain $G_{skew} < 0.01$ and saturating the manifold expansion to $\gamma_{max} = 2.0$.

## 10 Numerical Stability and Manifold Constraints

In the initial deployment of the **FAM-HOUSE** on the STM32F401, two distinct stability phenomena were observed during high-dynamic transitions and stationary states. These required the implementation of specific numerical "firewalls" within the SpFI system.

### 10.1 Asymmetric Mean-Shift and Yaw Instability

The most prominent artifact observed was a persistent yaw-axis drift exceeding 360°/s while the hardware was in a quasi-static state. This was traced to a mathematical decoupling between the *deformable manifold* and the *weighted mean reconstruction*.

When the SpFI system applied asymmetric scaling ($\gamma_i^+ \neq \gamma_i^-$) to a sigma point pair without a corresponding dynamic adjustment of the weights $W_m$, the geometric center of the point cloud shifted away from the true state estimate $\hat{x}_k$. In the context of a 7-state quaternion filter, this "mean shift" was erroneously interpreted by the Unscented Transform as a high-magnitude angular velocity. To mitigate this, we enforced a **Symmetric Manifold Constraint**:

$$\gamma_{i+1} = \gamma_{i+1+n} = \mathcal{F}(\zeta_i, \Omega_k) \tag{20}$$

where $\mathcal{F}$ is the fuzzy inference mapping. This ensures that even under manifold expansion, the point cloud remains centered on the predicted mean, eliminating artificial rotational drift.

### 10.2 Manifold Explosion and Quaternion Collapse

During high-innovation events (e.g., rapid 90° step inputs), the feedback loop between the point-wise distance $\zeta_i$ and the scaling factor $\gamma_i$ was susceptible to a "Manifold Explosion." If a single sigma point was propagated into a non-physical orientation space, the resulting NaN values in the innovation covariance $S_z$ led to a total filter collapse.

We addressed this by implementing an **Emergency Recovery Firewall** within the SpFI update loop. If any feedback parameter $\zeta_i$ registers as non-finite, the local manifold scaling for that dimension is force-reset to the nominal value ($\gamma = 1.0$). Furthermore, we constrained the maximum manifold expansion to $\Gamma_{\max} = 2.0$ to ensure that the "warped" sigma points remain within the valid linearization window of the quaternion exponential map.

### 10.3 State-Space Masking for Unobservable Biases

Finally, it was observed that applying fuzzy expansion to the gyro bias states $(b_x, b_y, b_z)$ increased the filter's sensitivity to high-frequency IMU noise. Since biases are slow-moving and unobservable through the accelerometer alone, we implemented *Manifold Masking*, where:

- **Dynamic States $(x, y, z)$:** Utilize full SpFI adaptive warping.
- **Parameter States $(b_x, b_y, b_z)$:** Are restricted to a nominal symmetric distribution.

This hybrid approach preserves the higher-order tracking capabilities of the FAM-HOUSE for orientation while maintaining the long-term stability of the bias estimation.

## 11 Experimental Results and Performance Analysis

The **FAM-HOUSE** architecture was validated using a 6-DOF IMU baseline. Initial testing revealed that the filter is highly sensitive to the correct mapping of rotational kinematics; an architectural inversion of the accelerometer and gyroscope vectors led to immediate divergence.

Upon rectification, the **SpFI system** demonstrated a proactive manifold expansion during high-G maneuvers. By dynamically adjusting the sigma point spread based on the **IMU Intensity** ($\Omega_k$), the filter maintained a lower **NIS** compared to the static SR-UKF. Specifically, the "Fuzzy Warping" allowed the filter to remain consistent during transient accelerations that typically cause standard UKFs to reject measurements as outliers.

### 11.1 Weight Normalization for Asymmetric Manifolds

To maintain the numerical stability of the Square-Root framework during asymmetric manifold deformation, we introduce **Dynamic Weight Scaling**. Since the SpFI system decouples

the displacement factors $\gamma_i^+$ and $\gamma_i^-$, the standard symmetric weights $W_c$ no longer satisfy the second-moment consistency of the Unscented Transform. We utilize a locally-compensated weighting scheme:

$$W_{c,i}^{\text{dyn}} = \frac{W_{c,i}^{\text{nom}}}{(\gamma_i^{\pm})^2} \tag{21}$$

This compensation ensures that the innovation matrix remains positive-definite even under extreme "warping" ($\gamma = 3.5$), preventing the Cholesky-factor collapse observed during high-transient maneuvers.

### 11.2 Real-Time Manifold Projection

To facilitate high-fidelity diagnostic monitoring, the 7-dimensional manifold is projected onto a 3D Euclidean space via a *Symmetric Difference Transformation*. The "bloom" or expansion of the sigma point cloud is driven by the coupling of the angular displacement $\theta_\delta$ and the fuzzy scaling factor $\gamma_i$. This visualization allows for the immediate identification of *filter over-confidence* or *innovation spikes*, as the cloud's volume serves as a real-time proxy for the determinant of the state covariance matrix $|S|$.

## 12 Computational Efficiency and Real-Time Performance

The **FAM-HOUSE** algorithm was benchmarked on an ARM Cortex-M4 (STM32F401) clocked at 84 MHz. Despite the introduction of the *Sigma Point Fuzzy Inference* (SpFI) system and a 15-point manifold, the filter achieved a stable update frequency of **800 Hz**.

This high-throughput performance is attributed to the integration of a custom Householder QR decomposition and the utilization of the ARM CMSIS-DSP v1.15.0 floating-point acceleration. With a total iteration latency of less than 1.25 ms, the FAM-HOUSE provides a deterministic state estimation loop suitable for the inner-loop attitude control of agile micro-aerial vehicles (MAVs), where high-bandwidth sensing is critical for rejecting non-Gaussian mechanical vibrations.

### 12.1 Dynamic Manifold Geometry and Visual Diagnostics

The core innovation of the **FAM-HOUSE** is visually demonstrated through the real-time projection of the sigma point cloud. Unlike the rigid, spherical symmetry of a standard SR-UKF, the **FAM-HOUSE** manifold exhibits "Elastic Deformation."

During steady-state operation, the cloud remains tightly clustered ($1.0 \leq \gamma \leq 1.1$), indicating high estimation confidence. However, during high-rate angular transients, the **SpFI system** induces an asymmetric "bloom." This elongation along the axis of rotation—coupled with the intensity-based color mapping—provides a direct visual representation of the filter's ability to adapt its internal state-space geometry to non-Gaussian noise distributions and local model uncertainties.

## 12.2 Fuzzy Membership Design

The adaptation of the manifold scaling factor $\gamma$ is governed by a piecewise linear membership function $\mu(\Omega_k)$, where $\Omega_k$ represents the normalized IMU intensity. To minimize the computational overhead on the Cortex-M4, we avoid Gaussian or Sigmoidal kernels in favor of a trapezoidal profile:

$$\gamma(\Omega_k) = \begin{cases} 1.0 & \Omega_k < \Omega_{min} \\ 1.0 + (\gamma_{max} - 1.0)\frac{\Omega_k - \Omega_{min}}{\Omega_{max} - \Omega_{min}} & \Omega_{min} \leq \Omega_k \leq \Omega_{max} \\ \gamma_{max} & \Omega_k > \Omega_{max} \end{cases} \tag{22}$$

where $\Omega_{min}$ and $\Omega_{max}$ define the "crossover" thresholds for nominal and extreme flight dynamics, respectively.

## 12.3 Dual-Channel Manifold Adaptation

The **SpFI** system is bifurcated into rotational ($\gamma_\omega$) and translational ($\gamma_a$) adaptation channels to decouple kinematic maneuvers from measurement noise. The orientation-state manifold is scaled as a function of the *Angular Rate Intensity*, while the bias-state manifold is modulated by the *Linear Acceleration Residual*:

$$\Gamma = \text{diag}(\gamma_\omega, \gamma_\omega, \gamma_\omega, \gamma_\omega, \gamma_a, \gamma_a, \gamma_a) \tag{23}$$

This decoupling ensures that the filter maintains a rigid bias estimate during high-rate rotations while simultaneously allowing for manifold expansion to mitigate the impact of non-gravitational accelerations during transient shocks.

# 13 Conclusion and Future Work

In this paper, we introduced the **Fuzzy Adaptive Manifold for Higher-Order Unscented Kalman Filters (FAM-HOUSE)**, a novel estimation framework that addresses the limitations of fixed-geometry sampling in highly nonlinear environments. By integrating the **Sigma Point Fuzzy Inference (SpFI)** system directly into the sigma point generation process, we have moved beyond simple covariance scaling toward a truly *deformable* uncertainty representation.

The key contributions of this research are summarized as follows:

- **Asymmetric Sampling:** The transition from global scaling to point-wise scaling ($\gamma_i^{\pm}$) allows the filter to capture higher-order moments, such as skewness and kurtosis, which are typically ignored by standard UKF and CKF implementations.
- **Proactive Adaptation:** By incorporating the IMU-derived Dynamic Intensity ($\Omega_k$) into the SpFI, the filter anticipates nonlinearities caused by high-dynamic maneuvers before they manifest as innovation errors.

- **Numerical Robustness:** The implementation within a Square-Root (SR) framework ensures that despite the introduction of asymmetric weighting, the covariance matrices remain positive-definite and computationally stable.

Experimental results demonstrate that the FAM-HOUSE significantly outperforms traditional estimators in scenarios involving rapid trajectory changes and non-Gaussian noise. The SpFI-driven geometry provides a more accurate grouping around the nonlinear manifold, leading to lower Mean Squared Error (MSE) and improved filter consistency.

### 13.1 Future Work

Future research will explore the automated optimization of the SpFI membership functions using evolutionary algorithms or reinforcement learning. Additionally, we aim to extend the FA-HOUSE to multi-sensor fusion environments where the SpFI can dynamically weight different sensor modalities based on their individual reliability and environmental interference.

## 14 Acknowledgments

The author(s) would like to acknowledge the use of Large Language Model (LLM) tools in the optimization of the square-root unscented Kalman filter (SR-UKF) algorithms, specifically in the translation of manifold kinematics to embedded C code and the development of the real-time Unity-based manifold visualization.